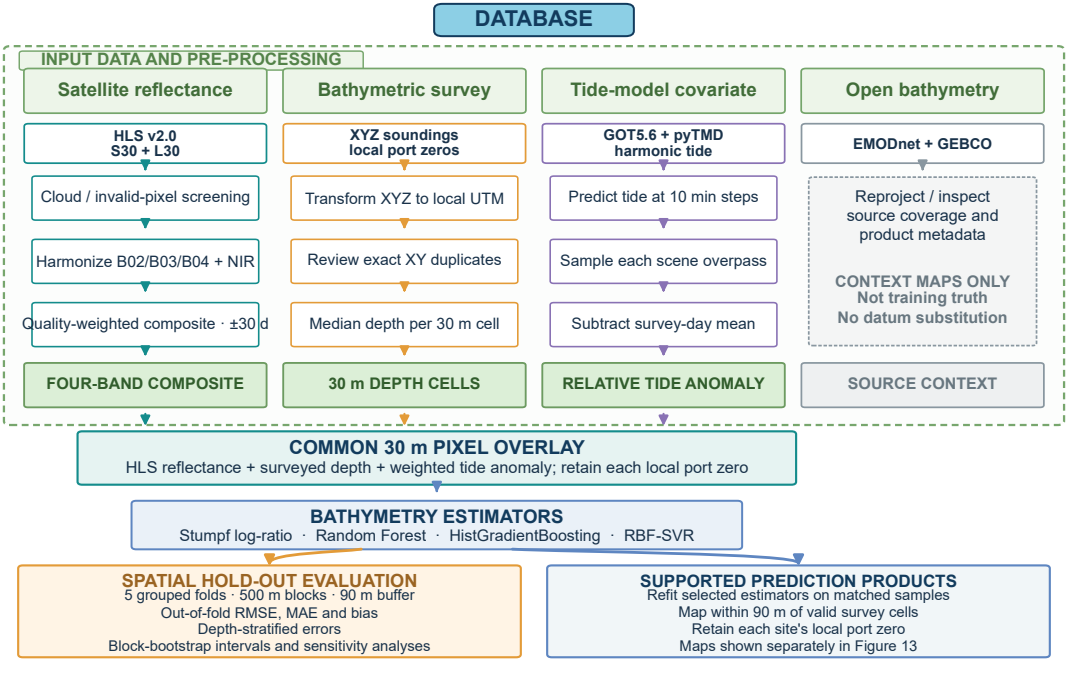

DATABASE
INPUT DATA AND PRE-PROCESSING
Satellite reflectance
Bathymetric survey
Tide-model covariate
Open bathymetry
HLS v2.0
S30 + L30
XYZ soundings
local port zeros
GOT5.6 + pyTMD
harmonic tide
EMODnet + GEBCO
Cloud / invalid-pixel screening
Transform XYZ to local UTM
Predict tide at 10 min steps
Reproject / inspect source coverage and product metadata
Harmonize B02/B03/B04 + NIR
Review exact XY duplicates
Sample each scene overpass
CONTEXT MAPS ONLY
Not training truth
No datum substitution
Quality-weighted composite · ±30 d
Median depth per 30 m cell
Subtract survey-day mean
FOUR-BAND COMPOSITE
30 m DEPTH CELLS
RELATIVE TIDE ANOMALY
SOURCE CONTEXT
COMMON 30 m PIXEL OVERLAY
HLS reflectance + surveyed depth + weighted tide anomaly; retain each local port zero
BATHYMETRY ESTIMATORS
Stumpf log-ratio · Random Forest · HistGradientBoosting · RBF-SVR
SPATIAL HOLD-OUT EVALUATION
5 grouped folds · 500 m blocks · 90 m buffer
Out-of-fold RMSE, MAE and bias
Depth-stratified errors
Block-bootstrap intervals and sensitivity analyses
SUPPORTED PREDICTION PRODUCTS
Refit selected estimators on matched samples
Map within 90 m of valid survey cells
Retain each site's local port zero
Maps shown separately in Figure 13